PROMPT FOR RESOLVING THE TWO-DIMENSIONAL COMPLEX JACOBIAN CONJECTURE

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Abstract. This document contains a full frontier-model research prompt for resolving the complex Jacobian conjecture in exactly two variables. The task is a universal polynomial-automorphism theorem for Keller maps $\mathbb{C}^2 \to \mathbb{C}^2$, or an explicit exact counterexample. Status snapshot: 20 July 2026.

1. Prompt

Current task statement

Let $P, Q \in \mathbb{C}[x, y]$ and define $F = (P, Q) : \mathbb{C}^2 \to \mathbb{C}^2$. Suppose the Jacobian determinant is a nonzero constant:

$$J(P, Q) = \det \begin{pmatrix} P_x & P_y \\ Q_x & Q_y \end{pmatrix} \in \mathbb{C}^\times.$$

After scaling one output coordinate, the constant may be normalized to 1. Resolve the two-dimensional complex Jacobian conjecture completely:

$$J(P, Q) \in \mathbb{C}^\times \quad \Longrightarrow \quad F \text{ is a polynomial automorphism of } \mathbb{C}^2?$$

The task is strictly dimension 2 over characteristic zero. Results or counterexamples in dimension 3 or higher, and results for the real Jacobian conjecture, do not settle this statement.

Complete-resolution standard

A complete resolution must prove one alternative:

- Affirmative alternative: prove that every polynomial Keller map $F : \mathbb{C}^2 \to \mathbb{C}^2$ has a polynomial inverse;
- Negative alternative: give explicit polynomials P, Q with exact coefficients and constant nonzero Jacobian but no polynomial inverse, together with an exact certificate of noninvertibility such as a collision or a rigorous algebraic obstruction;
- in either direction, prove every degree reduction, normalization, compactification, valuation argument, or ansatz is reversible and covers all possible two-variable maps.

What does not count

Partial progress is insufficient unless it logically implies the exact resolution above. In particular, the following do not count:

- a proof for bounded degree, special Newton polygons, triangular maps, symmetric maps, or any other proper subclass;
- a reduction to another unproved statement of equivalent difficulty;
- a locally invertible analytic map without a proof of global polynomial invertibility;
- a real counterexample, a map in dimension at least 3, or a rational map with denominators;
- a numerical near-constant Jacobian or approximate collision;
- a symbolic search that omits coefficient families without a theorem proving the omission harmless.

Use multiagent v2 aggressively and dynamically. You have up to 64 concurrent agents available. Do not use a fixed assignment such as "N agents for strategy X." Instead, manage the search using the following heuristics:

- Begin with a genuinely diverse portfolio of approaches. Agents should explore substantially different formulations, invariants, reductions, algebraic viewpoints, structural inductions, decompositions, optimization methods, exact computational searches, and formal-proof routes.
- Do not tell most agents the currently favored approach. Preserve independence during early rounds so that agents do not all converge to the same attractive but incomplete reduction.
- Maintain an explicit registry of approach families. Group agents by the mathematical mechanism they are using, not by superficial wording. If too many agents converge to one family, redirect some toward underexplored formulations.
- Do not allow one route to dominate merely because it gives elegant reductions. A route that ends at a missing lemma equivalent in strength to the original problem is not close to completion unless it supplies a genuinely new proof of that lemma.
- When an approach stalls at a theorem-strength missing lemma, mark that route as blocked. Reopen it only when an agent proposes a materially new mechanism, invariant, construction, or exact experiment.
- Keep several incompatible proof or construction routes alive through multiple rounds. Cross-pollinate ideas only after independent agents have developed them far enough to expose their real strengths and gaps.
- Use adversarial agents throughout. Every candidate theorem, construction, certificate, reduction, and program must be attacked independently. Assign separate agents to mathematical correctness, completeness of case analysis, numerical exactness, implementation soundness, and novelty checking.
- Require agents to return concrete lemmas, constructions, equations, counterexamples to proposed sublemmas, executable encodings, or formally stated proof obligations. Reject status reports, vague optimism, and claims that a global compatibility step is "routine."
- The root agent should repeatedly synthesize, challenge, redirect, and launch new rounds. Do not stop after the first wave fails. Convert promising patterns into exact conjectures and send those conjectures to independent proof and falsification teams.

Problem-specific search portfolio

In the initial independent rounds, ensure that the portfolio includes, but is not limited to, the following genuinely different families:

- Newton polygons, weighted degrees, leading forms, valuations at infinity, dicritical divisors, and compactifications of polynomial maps;
- algebraic geometry of etale endomorphisms of $\mathbb{A}^2$, field extensions, birationality, fibers, and behavior at infinity;
- polynomial automorphism structure, including Jung-van der Kulk theory, tame reductions, coordinate polynomials, and degree divisibility;
- minimal-counterexample arguments organized by degree pair, number of places at infinity, or topology of generic fibers;
- Groebner bases, resultants, elimination, Puiseux expansions, and exact symbolic computation used to expose universal identities or contradictions;
- collision-forced and factorized ansatzes for counterexample search, with the Jacobian equations solved exactly rather than over unconstrained raw coefficients;
- arithmetic reduction modulo primes, p-adic lifting, model-theoretic transfer, and effective bounds, with careful treatment of characteristic-zero recovery;
- topological, symplectic, differential, and dynamical reformulations only when the route back to polynomial invertibility is made exact.

Mandatory adversarial audit

Any surviving candidate must be checked against every item below by agents that did not originate the candidate:

- expand the Jacobian determinant symbolically and verify it is exactly a nonzero constant in the coefficient field;
- distinguish local analytic invertibility from global injectivity, surjectivity, birationality, and existence of a polynomial inverse;
- for a counterexample, verify all coordinates, collisions, field extensions, and noninvertibility arguments exactly over $\mathbb{C}$;

- for an affirmative proof, audit behavior at infinity and every exceptional divisor, branch, valuation, and degree case;
- check that no step silently assumes properness, injectivity, a real ordering, generic coefficients, or a nonvanishing leading term not guaranteed by the hypotheses;
- test each proposed general lemma against known polynomial automorphisms and known pathological polynomial maps that fail the constant-Jacobian hypothesis;
- formalize critical algebraic identities in a proof assistant or verify them independently in at least two computer algebra systems.

Required final artifact

The final response must contain a publication-level proof and exact supporting artifacts:

- for an affirmative solution, a complete proof applying to arbitrary $P, Q \in \mathbb{C}[x, y]$ with nonzero constant Jacobian;
- for a negative solution, the explicit polynomials in plain text and a computer-algebra file, plus exact verification of the determinant and noninvertibility;
- all computer-assisted eliminations, case splits, resultants, Groebner bases, or formal proofs in reproducible form;
- an independent symbolic verification report and cryptographic hashes;
- a section explaining precisely why higher-dimensional examples or reductions do not substitute for the two-dimensional argument.

Public search may be used for standard theorems on polynomial automorphisms, Keller maps, valuations, and published two-variable reductions. Do not search for a purported final proof or counterexample and do not answer merely that the plane case remains unresolved.

Do not return merely because current approaches fail or agents report theorem-strength gaps. Continue launching new rounds, reopening blocked approaches only when there is a genuinely new mechanism, and searching for fresh formulations.

Return only when a complete resolution has been found and survives adversarial audit. Do not return a reduction, a special case, an isolated missing lemma, an improved bound that leaves the target unresolved, a "best effort" literature review, or an explanation of why the problem is difficult.

Spend at least 8 hours of sustained search before even considering return or termination. If the execution environment imposes a hard stop before a complete resolution, do not fabricate success or elapsed work: preserve all machine-checkable artifacts and report only the strongest rigorously verified derivation together with the exact remaining gap.